Lab 05

Subscribing to APIs

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal, Admin Portal, Management Console, token endpoint).
- **`<GW_HTTPS_PORT>` / `<GW_HTTP_PORT>`** – API Gateway ports: 8243 (HTTPS) and 8280 (HTTP) plus any port offset.
- **`<APIM_HOME>`** – the folder where API Manager is extracted.

The servers use a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue.

Training Objective

Learn how to self-register in the Developer Portal, create an application, subscribe it to an API, and generate keys and access tokens.

Business Scenario

After PizzaShack successfully publishes its APIs, partners who would like to use the PizzaShack APIs as a base can open the Developer Portal, check its contents and subscribe to the API if interested.

High-Level Steps

- Browse the Developer Portal
- Define users via self-registration
- Subscribe to APIs

Part 1 – Browse the Developer Portal and Self-Register

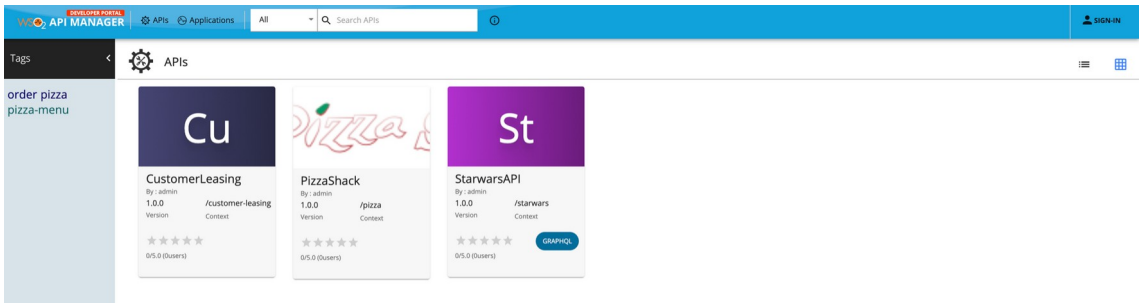
1. Open the Developer Portal. Published APIs are listed even before you sign in.

```
https://<APIM_HOST>:<HTTPS_PORT>/devportal
```

Example:

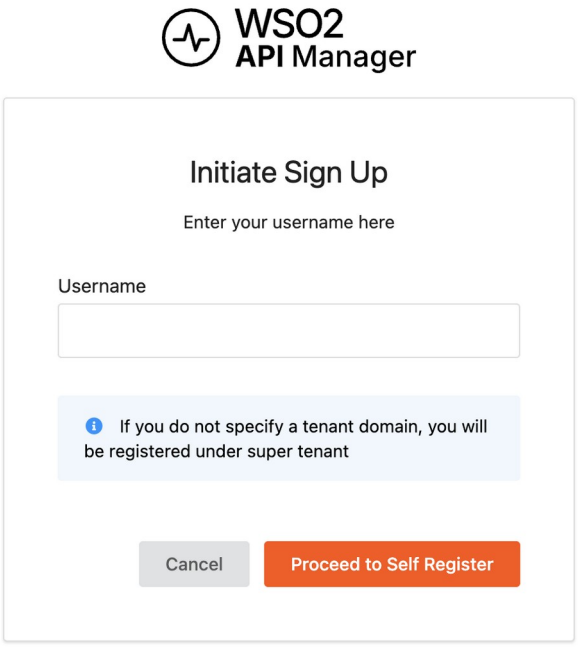
```
https://localhost:9443/devportal
```

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Users can register themselves in the Developer Portal (self sign-up is enabled by default).

2. Click **Sign-in** (top right) and then click **Create Account** on the login page.



3. Enter a user name and click **Proceed to Self Register**.

4. Fill in the required fields (first name, last name, email, password), accept the privacy policy and click **Register**.

   Note: self-registered users get the `Internal/subscriber` role automatically, so they can subscribe to APIs straight away.

5. Log in to the Developer Portal with the new user to check that self sign-up works, then log out.

   You can search APIs by their name, context or version, or by clicking the tags on the left.

   You can also try the API from the API console (**Try Out**), but you need a subscription and an access token first, which you get in the next steps.

Part 2 – Subscribe to an API

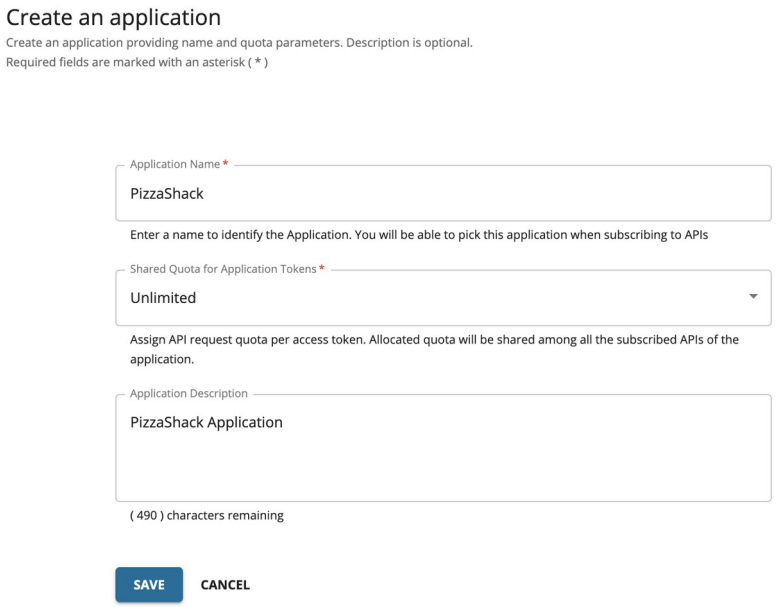
As a consumer, you can subscribe to an API by following these steps:

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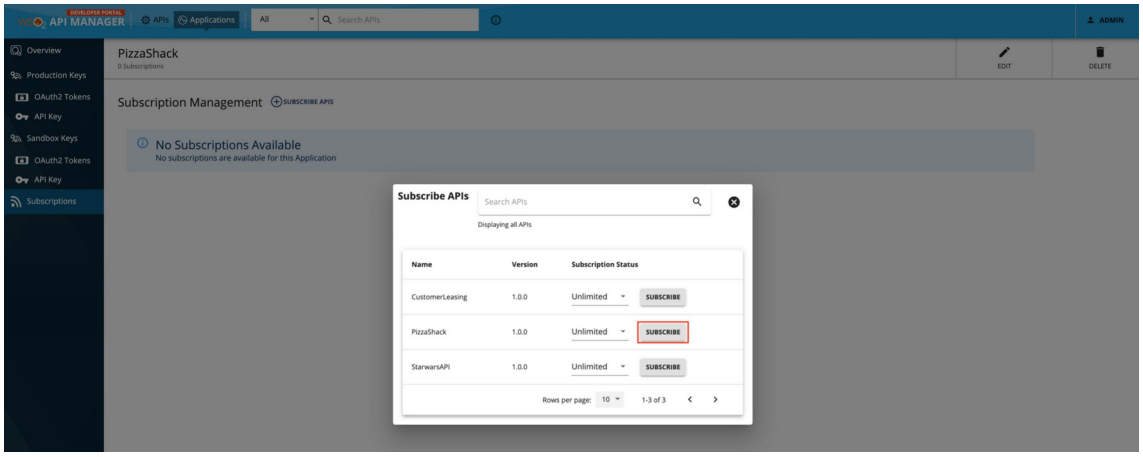
6. Log in to the Developer Portal as the **subscriber** user created in the Users and Roles lab. (A self-registered user also works, but the next lab uses subscriber, so use the same user here.) Logged-in users can also rate and comment on APIs.

7. Go to **Applications**, click **Add New Application**, enter the following information and click **Save**.

Field	Value
Application Name	`PizzaShack`
Shared Quota for Application Tokens	Unlimited
Application Description	PizzaShack Application



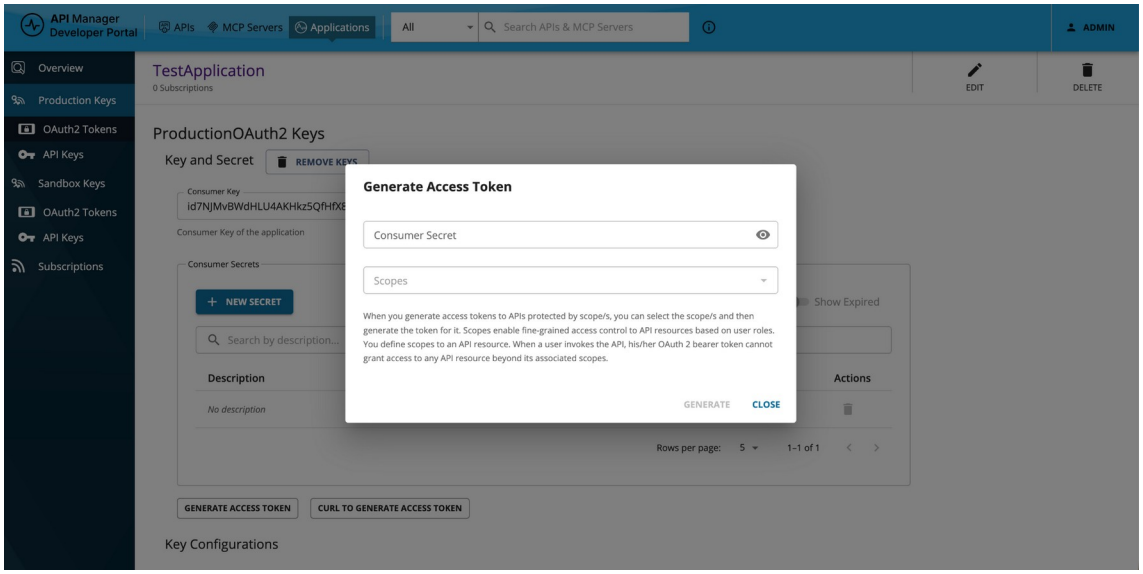
8. After the application is created, open its **Subscriptions** page, click **Subscribe**, find the **PizzaShack** API, choose the **Unlimited** business plan and click **Subscribe**.



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9. Go to **Production Keys → OAuth2 Tokens**. Under **Grant Types**, make sure **Password** and **Client Credentials** are selected (the PizzaShack web application in the next lab uses the password grant), then click **Generate Keys**.

10. Copy the **Consumer Key** and **Consumer Secret** and keep them for the next lab. In API Manager 4.7.0 the consumer secret is shown only once, when it is created. Then click **Generate Access Token**, paste the consumer secret, select the `order_pizza` scope and click **Generate**.



11. Copy the generated access token.

    You have now successfully subscribed to the API and can start using it.

Expected Outcome

A user has self-registered, and the **subscriber** user has an application (**PizzaShack**) that is subscribed to the PizzaShack API, with production keys and an access token that includes the `order_pizza` scope.

Optional: Multiple Consumer Secrets

WSO2 API Manager 4.7.0 supports several consumer secrets for one application at the same time. The main use is rotating the secret without downtime:

12. Generate a new secret while the old one is still active: on **Production Keys → OAuth2 Tokens**, click **Create New Secret**, optionally set a description and expiry, and copy the value (it is shown only once).

13. Move the clients to the new secret gradually. Access tokens can be generated with any valid secret.

14. Delete the old secret (**Delete Secret**) once nothing uses it.

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